

LIFE CYCLE ASSESSMENT-BASED DECISION FRAMEWORKS FOR NUTRIENT REMOVAL USING ENVIRONMENTALLY FRIENDLY SORBENTS IN TERTIARY WASTEWATER TREATMENT: A REVIEW

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Abstract. Sorbents-based technologies are increasingly being investigated as advanced solutions for the final removal of biogenic substances in tertiary wastewater treatment due to their ability to effectively remove phosphorus and nitrogen. Most of the existing scientific research focuses on the removal efficiency and physical and chemical properties of sorbents, while the environmental impact of sorbent-based systems throughout their life cycle is often assessed in a fragmented manner, using different methodological principles, making it difficult to compare the results obtained. This review systematizes and critically analyzes the current knowledge on the removal of biogenic substances using natural and modified sorbents, applies the methodological perspective of life cycle assessment (LCA), and moves from material property-oriented assessments to a structured, decision-supporting analytical approach. The main categories of sorbents are reviewed, taking into account the specifics of their application in tertiary wastewater treatment systems, and the essential life cycle stages, including sorbent production, transportation, operational use, regeneration processes and management at the end of their service life, are consistently assessed. Additionally, the most commonly used life cycle assessment methodological approaches are critically assessed, with particular attention to the selection of the functional unit, the definition of system boundaries and impact categories, such as global warming potential and eutrophication potential. Based on the synthesis of the reviewed studies, a conceptual, life cycle assessment-based decision-making system is proposed, integrating the efficiency of biogenic substances removal with environmental issues, aiming at a reasonable choice of technologies and their sustainable implementation. Finally, the main research gaps are identified, including the lack of harmonized methodological approaches to life cycle assessment, insufficient assessment of the efficiency of sorbent reuse and regeneration, and limited analysis of resource recovery; Recommendations for future research to base tertiary wastewater treatment on sustainability principles are also provided.

Keywords: life cycle assessment, sorbents, phosphorus removal, nitrogen removal, tertiary wastewater treatment, sustainability.

1. Introduction

Excess nitrogen (N) and phosphorus (P) discharged to receiving waters remain major drivers of eutrophication, triggering harmful algal blooms, oxygen depletion, and ecological degradation. The enrichment of nutrients is a consistent pressure in the Baltic Sea region that drives the need to have a nutrient management approach and enhanced performance of wastewater treatment (Hel-sinki Commission [HELCOM], 2009, 2023).

Within the European Union, regulatory frameworks require enhanced wastewater treatment and protection of sensitive waters. The Urban Waste Water Treatment Directive (Council of the European Communities, 1991)

sets requirements for collection and treatment, including enhanced treatment where needed, while the Water Framework Directive (European Parliament and Council, 2000) targets “good status” of European waters and strengthens nutrient control at catchment and wastewater treatment plant levels. As discharge limits tighten, conventional biological nutrient removal can be constrained when polishing to very low residual concentrations, increasing the relevance of tertiary steps.

Tertiary wastewater treatment, also referred to as advanced treatment or polishing, is the final treatment stage applied after conventional secondary treatment to further improve effluent quality before discharge to sensitive receiving waters or water reuse. Depending on the

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treatment objective, tertiary systems may target residual suspended solids, nutrients, pathogens, dissolved organic matter, or micropollutants. Broadly, tertiary treatment technologies can be grouped into: (i) physical separation processes, such as granular media filtration and membrane filtration; (ii) chemical and physicochemical processes, such as coagulation-precipitation, adsorption, ion exchange, and advanced oxidation; and (iii) biological polishing processes, such as post-denitrification and biofiltration. Within this broader technological landscape, sorbent-based nutrient removal belongs primarily to the physicochemical polishing group, especially adsorption-, ion-exchange-, and precipitation-assisted media systems, and is particularly relevant when very low residual phosphorus and ammonium concentrations must be achieved. Tertiary treatment is widely described as the final polishing stage after secondary treatment, and recent reviews likewise group advanced options into physical, chemical/physicochemical, membrane, oxidation, and biological polishing approaches.

Within this tertiary-treatment framework, sorbent-based nutrient removal has received increasing attention. Ammonium and phosphate can be removed using natural minerals (e.g., clinoptilolite rich zeolites), carbonate, carbonaceous (e.g., biochar) and engineered sorbents through ion exchange, surface complexation, and precipitation. Nevertheless, a significant portion of the literature focuses on adsorption capacity and removal efficiency, whereas life-cycle concerns of production, modification, regeneration, transport, and end-of-life are discussed in a haphazard manner.

The performance of the sorbents has to be considered in addition to the life-cycle burdens to guarantee that the benefits of the net sustainability are achieved at scale with the help of environmentally friendly sorbents. LCA offers a standardized methodology (International Organization for Standardization [ISO], 2006a, 2006b), yet the results of LCA of wastewater are subject to different functional units, functional boundaries, inventory assumptions, and choice of life cycle impact assessment (LCIA), which constrains comparability and may be biased towards technology (ISO, 2006a, 2006b; Corominas et al., 2013; Renou et al., 2008). This review summarizes tertiary nutrient removal using sorbents in the LCA point of view and suggests a concise choice structure that relates performance to the environmental effects in an effort to select decisions that are relevant to the decision maker.

2. Methodology

2.1. Review design and scope

This paper adopted a structured literature review approach to synthesize and critically evaluate published studies on sorbent-based nutrient removal in tertiary wastewater treatment from a life cycle assessment perspective. The review focused on natural, modified, and composite sorbents used for phosphorus and/or nitrogen removal, with particular attention to life cycle stages and methodological choices affecting environmental evaluation, including sorbent production, transport, operational use, regeneration, end-of-life management, functional unit selection, system boundaries, impact categories, and allocation assumptions. The synthesis was designed to support the subsequent discussion of sorbent categories, LCA considerations, sustainability hotspots, and decision-support implications presented in the manuscript.

2.2. Literature search strategy and study selection

The literature search was conducted as a structured review of peer-reviewed and institutional literature relevant to sorbent-based nutrient removal in tertiary wastewater treatment from a life cycle assessment perspective. To capture both technological and methodological studies, the search was performed using Scopus, Web of Science, and Google Scholar, covering publications from 2000 to 2024. This timeframe was selected to include both early foundational work on sorbent-based nutrient removal and more recent studies addressing life cycle assessment, sustainability evaluation, regeneration, and decision-support implications.

2.3. Research questions and thematic synthesis

The review was guided by the following research questions:

(1) Which sorbent categories and nutrient-capture mechanisms are most commonly reported for tertiary wastewater treatment? (2) Which methodological choices in life cycle assessment most strongly influence the environmental evaluation of sorbent-based systems? (3) Which life cycle stages and operational assumptions act as the main sustainability hotspots? (4) How can the available evidence be translated into a practical decision-support framework for selecting sorbent technologies?

After selection, the studies were analyzed thematically according to their contribution to: (i) sorbent categories and dominant removal mechanisms, (ii) methodological issues in LCA application, (iii) key sustainability hotspots, (iv) implications for technology selection, and (v) research gaps for future work. This thematic synthesis

provided the basis for the conceptual LCA-based decision framework proposed in the manuscript. The core characteristics of the studies included in the literature synthesis are summarized in Table 1.

Table 1. Core characteristics of the reviewed studies included in the literature synthesis

Study	Main focus	Relevance
Hedström (2001)	Zeolites for ammonium removal	Supports the discussion of natural mineral sorbents, ion exchange, and regeneration-related burdens
Karageorgiou et al. (2007)	Calcite for phosphate removal	Supports the discussion of natural carbonate-based sorbents and adsorption-precipitation mechanisms
Ahmad et al. (2014)	Biochar as sorbent platform	Provides background on biochar properties, feedstock dependence, and sorbent functionality
Almanassra et al. (2021)	Phosphate removal by biochars	Supports the discussion of phosphate-capture mechanisms and modification-related performance
Verma and Nadagouda (2021)	Graphene-based composites for phosphate removal	Supports the discussion of modified/composite sorbents and possible material and energy burdens
Corominas et al. (2013)	LCA applied to wastewater treatment	Provides the methodological basis for functional units, system boundaries, allocation, and reporting transparency
Renou et al. (2008)	Influence of LCIA method choice	Supports the discussion of method sensitivity and comparability of environmental results
Morelli et al. (2018)	Eutrophication models in LCA	Supports the discussion of uncertainties in eutrophication impact modelling
Lam et al. (2020)	Nutrient recycling from wastewater	Supports the treatment of recovery credits, multifunctionality, and scenario sensitivity
Roberts et al. (2010)	Biochar systems and climate/energy burdens	Supports the hotspot analysis related to upstream production energy and co-product assumptions

3. Sorbent-based nutrient removal in tertiary treatment

Usually, sorbent-based polishing is used as either fixed-bed filters, packed columns, or reactive media beds to eliminate remaining $\text{NH}_4^+/\text{NH}_3\text{-N}$ and $\text{PO}_4\text{-P}$ following biological treatment. Some of the major design drivers are empty-bed contact time, bed life/breakthrough,

headloss and backwashing requirements, and matrix effects (pH, alkalinity, competing ions), which also have an effect on regeneration frequency and life-cycle effects.

3.1. Natural mineral sorbents

Zeolite have high cat-ion exchange capacity with selectivity in NH_4^+ . The process is taken up mainly through ion exchange with $\text{Na}^+/\text{K}^+/\text{Ca}^{2+}$, regeneration is routinely performed with NaCl brine and the process allows a repetitive cycle, but places a burden on brine production and management (Hedström, 2001). Phosphate adsorption and Ca-phosphate precipitation take place under positive pH/saturation parameters; phosphate adsorption is pH-dependent and tends to be a combination of adsorption-precipitation response (Karageorgiou et al., 2007).

3.2. Biochar-based sorbents

Pyrolysis is used to convert biomass residues into biochar; its functionality is determined by the feedstock, temperature, ash/mineral content, functional group and porosity. In general, biochar is extensively examined as a universal sorbent platform, whereas the removal of phosphates is frequently enhanced significantly by mineral/metal enrichment (Ahmad et al., 2014). An overview of phosphate removal by biochar-based materials suggests capture processes such as electrostatic attraction (pH-dependent), inner-sphere complexation, and mineral precipitation, further favored by control of surface chemistry/mineral phases (Almanassra et al., 2021).

3.3. Modified/composite sorbents

Metal loading increases phosphate removal by complexing on metal (hydr)oxides and/or by precipitating metal-phosphate phases. Although performance may improve, modification can add to life-cycle burdens (reagents, energy) and also add leaching considerations, which need to be explicitly considered in LCA-based comparisons (Almanassra et al., 2021).

Graphene oxide (GO)/graphene composites would have the capability to expand the surface area and adsorption sites, which usually need functionalization to high phosphate affinity. Such materials can be either resource/energy demanding to manufacture; hence, their sustainability assertions must be validated via LCA as opposed to basing their assertions solely on performance in adsorption (Verma & Nadagouda, 2021).

3.4. Dominant capture mechanisms

In sorbent classes, the primary mode of nutrient removal is through: The rates of ion exchange (i), adsorption

(ii), pH-dependent (iii) inner-sphere complexation (P on Fe/Al sites), and precipitation (iv) all occur on the P site. These processes have effects on operational inputs (regeneration chemicals, pH control), as well as on end-of-life paths (reuse vs. disposal), and thus have a significant impact on life-cycle effects.

4. LCA considerations for sorbent-based tertiary polishing

LCA is particularly relevant for sorbent systems because upstream burdens (energy-intensive production) (Ye et al., 2025) and downstream burdens (regeneration chemicals, spent media handling) can undermine “environmentally friendly” claims. Wastewater LCA literature emphasizes strong sensitivity to methodological choices and the need for transparent reporting (Corominas et al., 2013).

4.1. Functional unit (FU)

A common use of treated 1 m³ is however misleading when the influent load and targets are not equal. To achieve tertiary polishing, comparability is enhanced when FU reflects a compliance or pollution removal, i.e., 1 m³ effluent within set N/P ranges or 1 kg P removed/1 kg N removed, depending upon the decision (Corominas et al., 2013).

4.2. System boundaries

For decision support, a cradle-to-grave scope is preferred: production/activation/modification; transport; operation (pumping/headloss, backwash, chemicals); regeneration loops (chemical manufacture and handling); end-of-life (landfill/incineration/stabilization/reuse); plus any recovery credits. One of the largest sources of non-comparability is omission/inconsistent inclusion of consumables and residuals (Corominas et al., 2013).

4.3. Impact categories and LCIA method choice

Core categories for nutrient-polishing systems include climate change, global warming potential (GWP), eutrophication potential (state freshwater vs. marine), and energy demand; acidification is often a supporting category. LCIA method choice can change ranking (Renou et al., 2008). The ranking may vary according to LCIA method choice (Renou et al., 2008). Eutrophication is method-sensitive specifically; a critical review shows discrepancies/shortcomings in models of eutrophication using LCIA (Morelli et al., 2018). For consistency in the European context, the selection of impact assessment methods may follow the International Reference

Life Cycle Data System (ILCD) recommendations developed by the Institute for Environment and Sustainability (IES) of the European Commission's Joint Research Centre (JRC).

4.4. Multifunctionality and credits

In situations whereby the fertilizer is replaced with spent sorbents or eluates, or wastes are used in the form of feedstock, allocation and credits may be dominant. The ISO instructions include the fact that allocation in case of system expansion/subdivision should be avoided and instead should be handled using clear rules. Whether it can be considered a critical review of LCA to wastewater-based nutrient recycling, it is important to note that recovery credits are scenario-volatile and have to be conducted with the help of sensitivity analysis (Lam et al., 2020).

5. Key sustainability hotspots

The life cycle burns that characterize energy and chemical inputs can make overall sustainability to dominate even when sorbents have high nutrient removal processes. This is usually the production energy footprint that often drives the carbonaceous sorbents since pyrolysis/activation energy requirement and local electricity mix is heavily impacting the climate-change and energy indicators; biochar LCAs demonstrates that assumptions about co-product energy recovery can sharply change the findings (Roberts et al., 2010). Chemical modification may be burdensome there may be effects of reagent manufacture, composite preparation, therefore should be reported on functional units consistent with the treatment service (e.g., per kg P removed) not per gram of media.

There may be also a material dependence of impacts on transport and replacement frequency when transporting large distances over long distances or requiring frequent replacement of a medium and bed life can become matters of supply-chain choices (Corominas et al., 2013). Regeneration vs. disposal decisions further turn the outcomes: regeneration can cause a decrease in the virgin media demand, however, it involves periodic chemical and energy use and must have side-stream management; it is only good when chemicals do not add up a lot and when regeneration will work efficiently after numerous cycles. Since sorbent polishing moves the pollutants in water to the solids the residual and final management should be considered with realistic options of fate (stabilization, landfill, incineration or controlled reuse). Lastly, nutrient recovery credits (e.g., fertilizer substitution) can have a powerful effect but are

instance-specific and hypothesis-dependent, thus, the findings must be reported with and without credits and analysed by sensitivity analysis (Lam et al., 2020).

6. Proposed LCA-based decision framework for selecting sorbent technologies

The process of identifying environmental trade-offs in the tertiary removal of nutrients using sorbent would require life cycle assessment (LCA) but this does not mean that the results of an LCA is intrinsically accurate in determining the best technology to use since the actual decisions rely on compliance reliability, cost, and implementability. Hence, the small decision framework will prove beneficial to screen LCA results in a clear prioritization, as well as to avoid the situation when an unacceptable choice (e.g., an effluent non-compliant) is taken based on another measure.

The suggested structure has a two-step logic. Based on Stage A (feasibility/compliance), the elimination of alternatives that fail to satisfy non-negotiable requirements precedes any scoring. Characteristic screening test consists of; capability of meeting necessary effluent target of $\text{PO}_4\text{-P}$ and $\text{NH}_4\text{-N}$ with a realistic wastewater matrix; approved risk of leaching/contaminant release, especially where metal-modified sorbents involved; conformability to existing hydraulic and footprint (e.g., headloss, backwashing requirements, retrofit limitations); and regulatory acceptability of end-of-life pathways of spent media (reuse/land use limitations, stabilization, and landfill or incineration).

In Stage B, the remaining viable options are evaluated using multi-criteria decision analysis (MCDA), based on a weighted scoring matrix where the weights are based on the priorities of the stakeholders so that they can explicitly make trade-offs and these are not implied. The scoring criteria may be divided into four pillars which have quantifiable measures; treatment performance (effluent $\text{PO}_4\text{-P}$ and $\text{NH}_4\text{-N}$, breakthrough/bed life, and regeneration retention across cycles when applicable), environmental performance based on LCA results (small set which includes climate change/GWP, different eutrophication potential with freshwater or marine clearly defined, energy demand), techno-economic performance capital expenditure (CAPEX) or operational expenditure (OPEX) proxies and, where available, cost per m^3 polished or cost per kg P removed), and practical feasibility (the functional unit, system boundary, inventory assumptions and LCIA method) used in the studies should be reported transparently in accordance with ISO 14040:2006 and ISO 14044:2006 to improve the comparability of LCA results. CIA weights can be determined either by stakeholder elicitation or by the

Analytic Hierarchy Process (AHP) that gives traceable pairwise ratings and a consistency check (Saaty, 1980). Since indicators are measured in different units they should be normalized (e.g., by min-max scaling to a 0–1 scale, inverting indicators with a lower-is-better meaning, such as GWP and cost). It can then be ranked using transparent weighted sum model and robustness may be commonly determined using TOPSIS provided there is space (Hwang & Yoon, 1981).

Lastly, due to the uncertainty in media lifetime, regeneration efficiency degradation, effects of real wastewater matrix, and feasibility of end of life, the framework must report a minimum of robustness set: scenario analysis (single-use versus regenerated; landfill versus reuse; local versus imported supply chains), weight sensitivity (e.g., $\pm 20\%$), and key parameter sensitivity (production energy intensity, regeneration chemical dose, and media lifetime). Outputs are decision-useful and presentable at a conference: a ranked shortlist to a specified polishing objective, a contribution profile of what criteria is driving the best option(s) and a brief decision map to review and practitioners to summarize trade-offs.

7. Research gaps and future directions

Key gaps limiting decision-ready sustainability assessment include:

1. Harmonization: inappropriate FU and boundary decisions; FU, boundary, allocation and assumptions should be reported in the studies in accordance with ISO 14040:2006 and ISO 14044:2006 (Corominas et al., 2013).
2. Inventory at field scale field/pilot is more realistic in terms of real operational demands (headloss/backwash energy, variable influent, realistic bed life).
3. Integration with economics: must include at least cost per m^3 polished or per kg removed and consumables sensitivity.
4. Regeneration/recovery realism: model regeneration loops represent overt scenarios, not presumed benefits, and recovery credits (Lam et al., 2020).
5. Decision-support translation: MCDA technology and scenario investigation can be used to achieve greater acceptability when used by utilities and regulators.

8. Conclusions

A potential future of the tertiary polishing is sorbent-based, which is based on the principle of developing a remarkably stricter level of nitrogen and phosphorus discharge limits, yet the validity of the term

environmentally friendly should be justified not by performance on a removal basis. Following the LCA as an element of a systematic decision-making program enhances accountability and discourages deceptive judgment based on narrow scopes of the system under consideration and inconsistent assumptions. The two-step approach is a pragmatic approach: feasibility/compliance screening and MCDA ranking. This gives, in addition to treatment performance, cost and implementation restrictions, the opportunity to evaluate environmental impacts (e.g., GWP, eutrophication potential, energy demand). With this sort of approach, easily comprehensible deployment-relevant rankings are generated, which can be examined by reviewers, utilities, and regulators. The key research requirements are field scale realistic energy and chemical and media-lifetime inventories. LCA choices need to be a reported and harmonized (functional unit, boundaries, LCIA method, and allocation) in order to be comparable. Scenario and sensitivity analysis of regeneration, reuse, and end-of-life pathways with side-stream management need to be explicitly modelled. Lastly, coupled LCA-cost reporting should be done to facilitate reasonable, feasible technology choices in upgrading tertiary nutrient removals.

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**GYVAVIMO CIKLO VERTINIMU GRINDŽIAMŲ
SPRENDIMŲ MODELIAI BIOGENINĖMS
MEDŽIAGOMS ŠALINTI NAUDOJANT APLINKAI
NEPAVOJINGUS SORBENTUS TREČINIO NUOTEKŲ
VALYMO METU: APŽVALGA**

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Santrauka. Sorbentais pagrįstos technologijos vis dažniau tiriamos kaip pažangūs biogeninių medžiagų galutinio šalinimo sprendimai trečiojo nuotekų valymo etape dėl gebėjimo efektyviai pašalinti fosforą ir azotą. Dauguma mokslinių tyrimų daugiausia dėmesio skiria šalinimo efektyvumui bei sorbentų fiziniams ir cheminėms savybėms, o sorbentais pagrįstų sistemų poveikis aplinkai viso gyvavimo ciklo metu dažnai vertinamas fragmentiškai, taikant nevienodus metodinius principus. Todėl gautus rezultatus sunku palyginti. Šioje apžvalgoje sisteminamos ir kritiškai analizuojamos dabartinės žinios apie biogeninių medžiagų šalinimą naudojant natūralius ir modifikuotus sorbentus, taikant gyvavimo ciklo vertinimo metodologinę perspektyvą. Pereinama nuo į medžiagų savybes orientuotų vertinimų prie struktūruoto analitinio požiūrio, galinčio pagrįsti sprendimų priėmimą. Apžvelgiamos pagrindinės sorbentų kategorijos ir į jų taikymo ypatumai trečiojo nuotekų valymo sistemose, taip pat nuosekliai vertinami svarbiausi gyvavimo ciklo etapai: sorbentų gamyba, transportavimas, eksploatacinis naudojimas, regeneracija ir tvarkymas pasibaigus jų naudojimo laikotarpiui. Papildomai kritiškai įvertinamos dažniausiai taikomos gyvavimo ciklo vertinimo metodinės priemonės, ypatingą dėmesį skiriant funkcinio vieneto parinkčiai, sistemos ribų apibrėžčiai bei poveikio kategorijoms, tokioms kaip visuotinio atšilimo potencialas ir eutrofikacijos potencialas. Remiantis apžvelgtų tyrimų sinteze, siūloma konceptuali gyvavimo ciklo vertinimu pagrįsta sprendimų priėmimo sistema, integruojanti biogeninių medžiagų šalinimo efektyvumą su aplinkosauginius aspektus, siekiant pagrįsto technologijų pasirinkimo bei tvaraus jų diegimo. Galiausiai identifikuojamos pagrindinės mokslinių tyrimų spragos, įskaitant suderintų gyvavimo ciklo vertinimo metodologinių priemonių trūkumą, nepakankamą sorbentų pakartotinio naudojimo ir regeneracijos efektyvumo vertinimą bei ribotą išteklių atgavimo analizę. Taip pat pateikiamos rekomendacijos būsimiesiems tyrimams, skirtos trečiojo nuotekų valymo sprendimų plėtrai remiantis tvarumo principais.

Reikšminiai žodžiai: gyvavimo ciklo vertinimas, sorbentai, fosforo šalinimas, azoto šalinimas, trečiasis nuotekų valymas, sprendimų modeliai, tvarumas.